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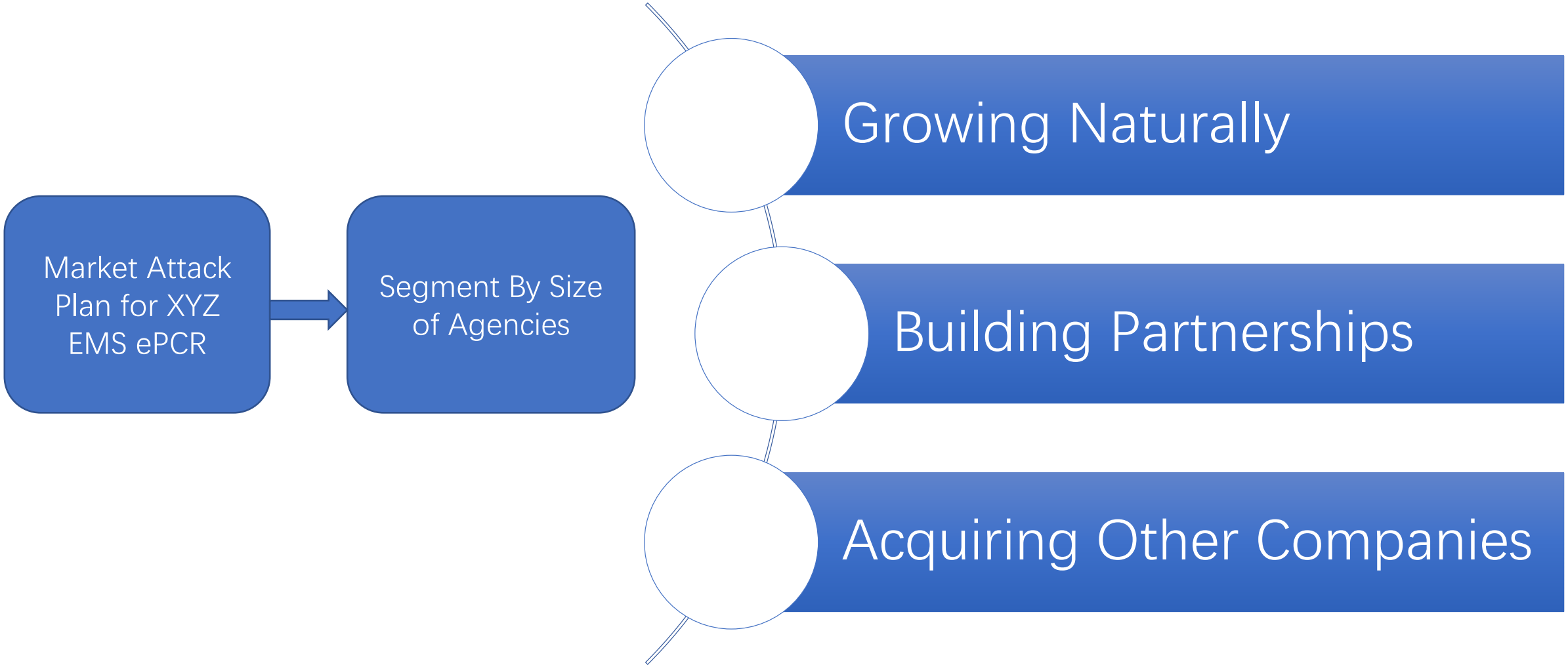
Company B – Opportunities for Division XYZ EMS in ePCR market

October, 22nd, 2019

Group 4: Soham Modi, Frances Wang, Tonia Jin, Jake Ashcraft

Executive Summary



Contents

Situation & Introduction

- Situation & Complication
- Objectives

Analysis

- Market Size
- Segmentation

Recommendations

- Market Entry Analysis and Risks
- Implementation Initiatives

Competitive Advantages of Company B

- **global leader** in medical technology
- provides **a wide range of products and therapies**, including an XYZ EMS ePCR system.
- XYZ EMS **meets the growing demand** for reliable and complete patient record data

XYZ EMS is sold to **only a select segment** of the pre-hospital EMS environment in the U.S.

Segment the Market for XYZ EMS ePCR

What opportunities are available for Company B to combine both internally generated data and external market research in order to **construct a better market segmentation** for their new products?

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Total number of EMS agencies:

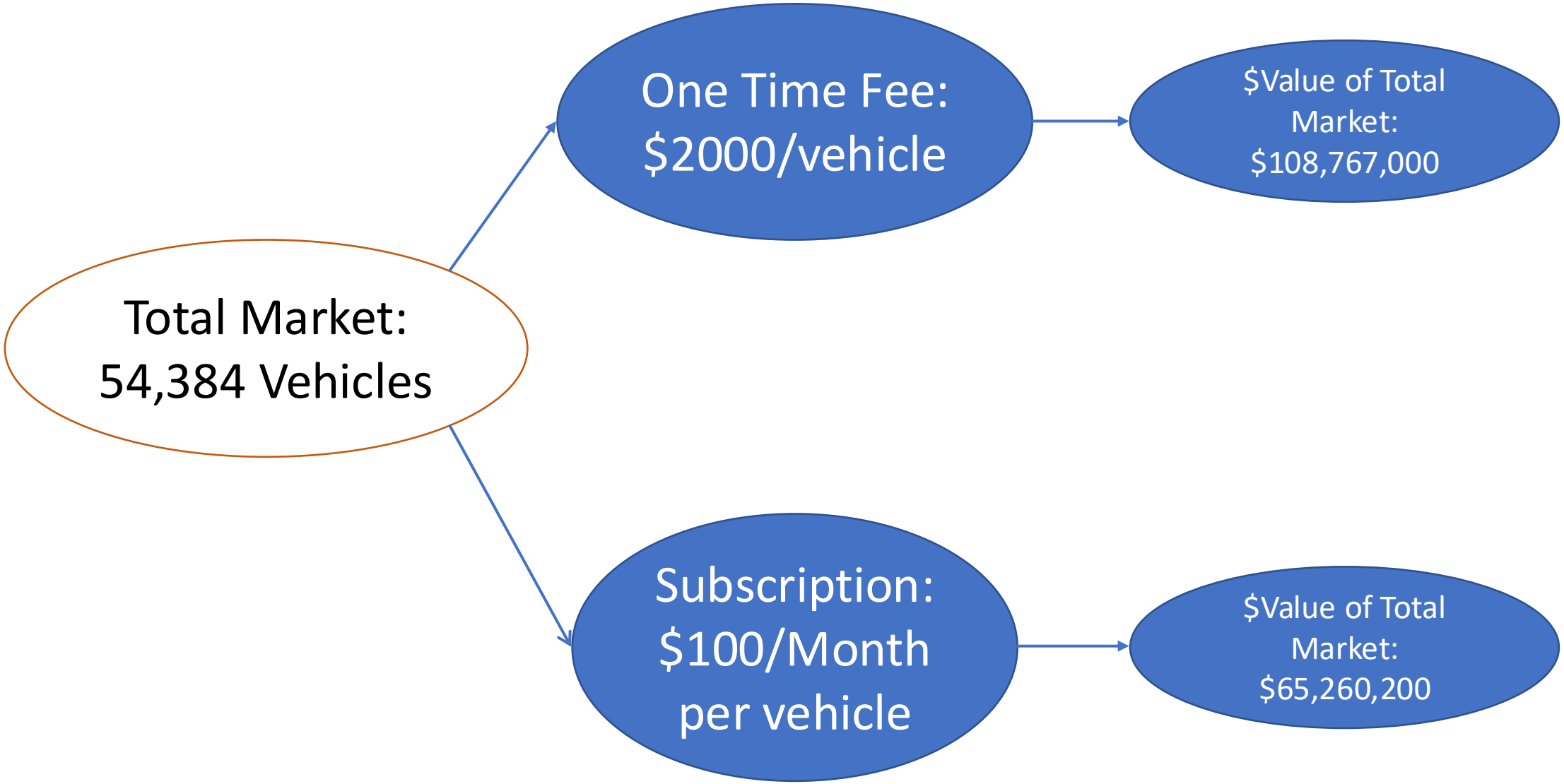
19,437

Estimated Total number of Vehicles:

54,384

Data Resource

1.Based on extrapolation from geographical breakdown survey for 800 EMS centers. “Characterizing Local EMS Systems”, U.S. department of transportation, National Highway Traffic safety Administration, 2013.
2.National Association of State EMS Officials, “National EMS assessment 2011”
3.Based on extrapolation from NYS data by county. “Listing of Ambulance and Advanced Life Support First Response Services in New York “ 2015.


Total Market:
54,384 Vehicles
One Time Fee:
$2000/vehicle
$Value of Total Market:
$108,767,000
Subscription:
$100/Month
per vehicle
$Value of Total Market:
$65,260,200

How should Company B segment the market?

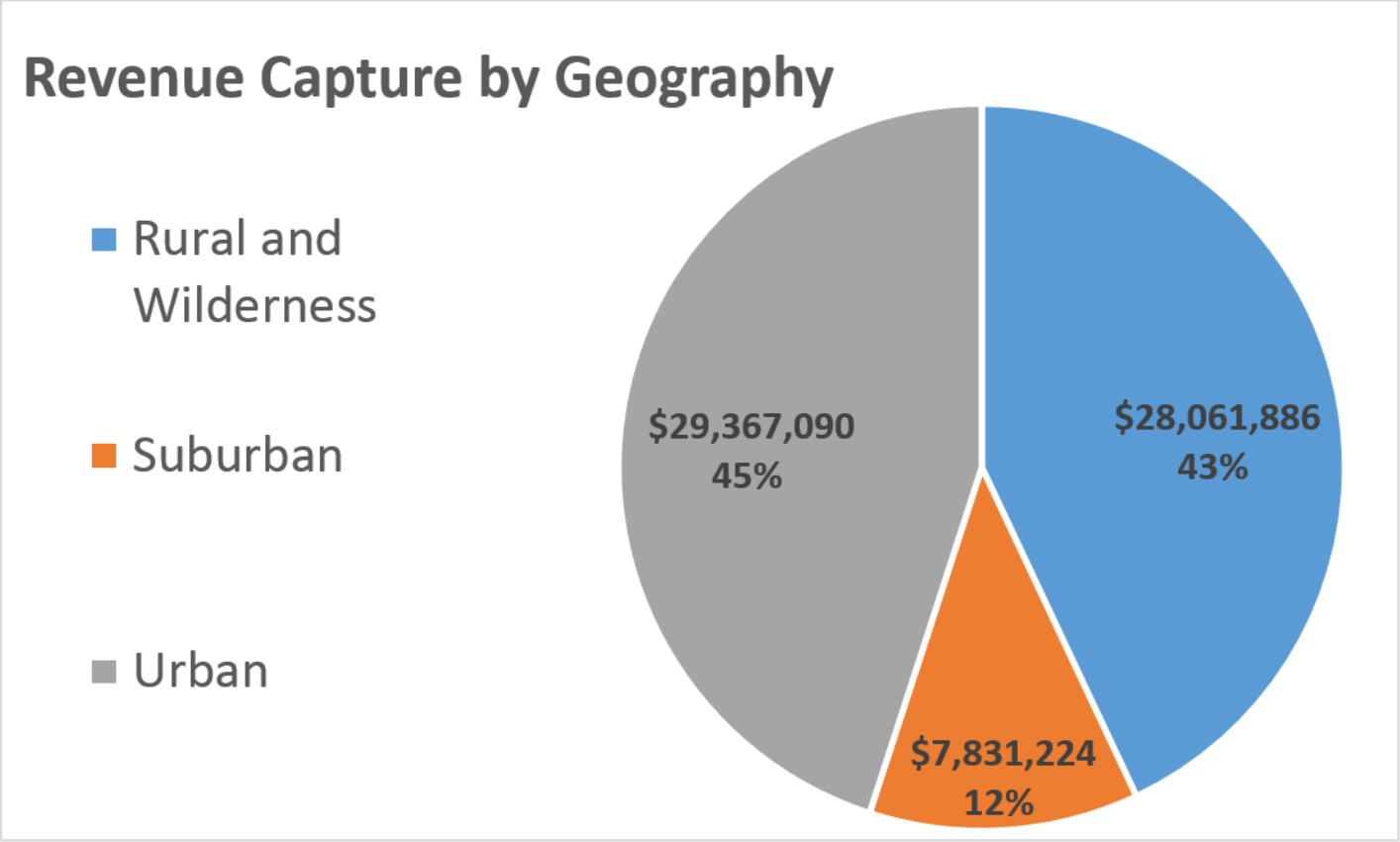
Geography

- Rural and Wilderness
- Suburban
- Urban

No. of Vehicles/Agency

- 1-2
- 3-6
- 7-10
- 11-15
- >15

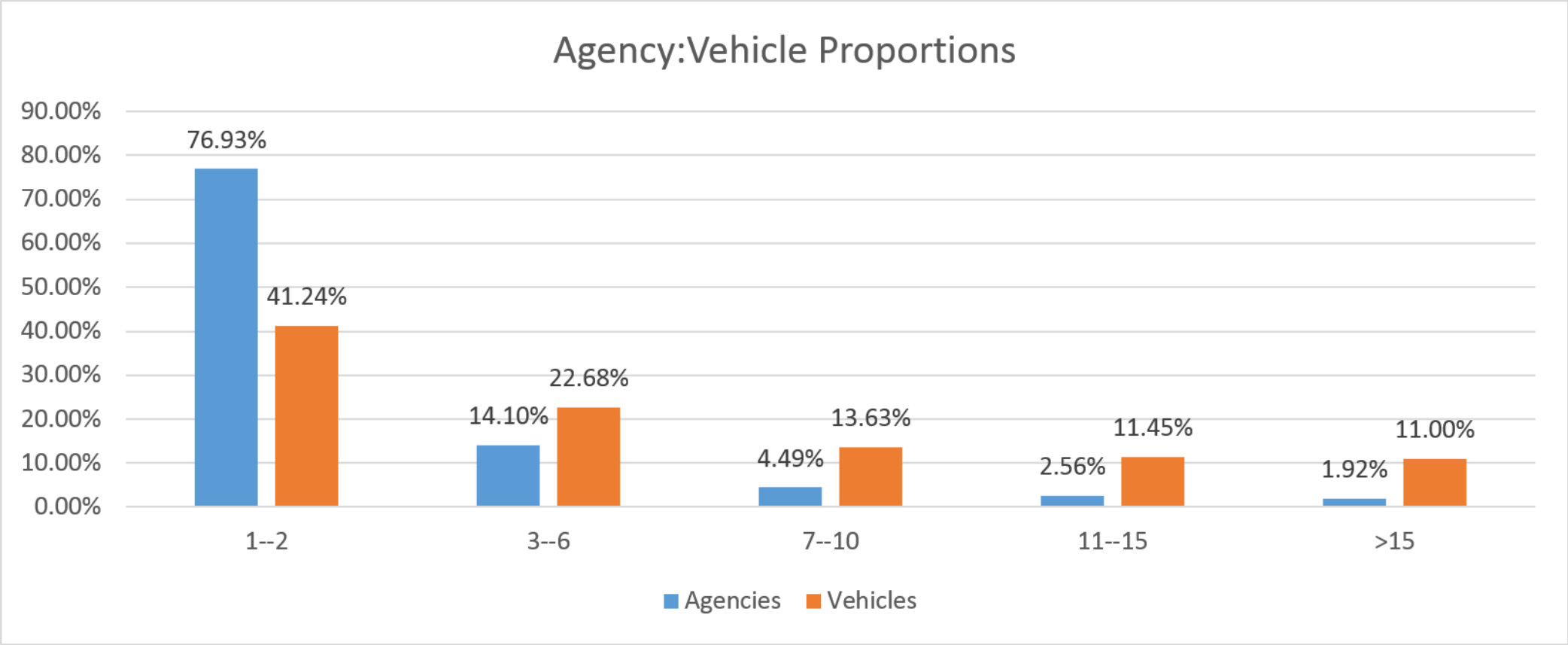
How can geography be used to determine a target market?



Data Resource

- Based on extrapolation from geographical breakdown survey for 800 EMS centers. “Characterizing Local EMS Systems”, U.S. department of transportation, National Highway Traffic safety Administration, 2013.
- National Association of State EMS Officials, “National EMS assessment 2011”

How can number of vehicles be used to determine a target market?



Data Resource

- National Association of State EMS Officials, "National EMS assessment 2011"
- Based on extrapolation from NYS data by county. "Listing of Ambulance and Advanced Life Support First Response Services in New York " 2015.

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Agencies with over 10 vehicles can be targeted with in house trainings, classroom instructions and software interactions leveraging applications on ePCR in Urban areas

ePCR

A1 Business and non-business environment

- **Competitive market** with several large and medium sized incumbents
- **High buyer power** – market driven by EMS agencies
- **Favorable societal and regulatory** environment for new information systems

A2 Market size

- High market growth (expected to reach USD 311.2 million by 2026, with 8.40% CAGR): **enables enhancement of the process to increase revenue for the clients is anticipated to drive the market**
- **Raising demand**

A3 Fit with internal capabilities

- Company B has **strong innovation capability**
- **Innovation from other related sectors** (e.g., tech, healthcare) can be applied to **ePCR**
- **Weakness in access to customers**

Factors Impacting Usability

B1 Trainings

- Based on survey result, EMS practitioners who received no training on their ePCR rated the ePCR usability **much lower than those who had received training**
- Those who received interactive training had **the highest usability score**

B2 Data Quality

- Few EMS agencies have designated a specific **level of proficiency** for ePRC use
- Clients are not paying enough attention on **quality data collection**
- Requires accuracy, timeliness, completeness, relevance, reliability and consistency

B3 Integrating System with Diverse Functions

- Clients have **high expectation** on diversed functionalities
- **Usability** is the key

Chosen segments

Agency Size: > 6 vehicles

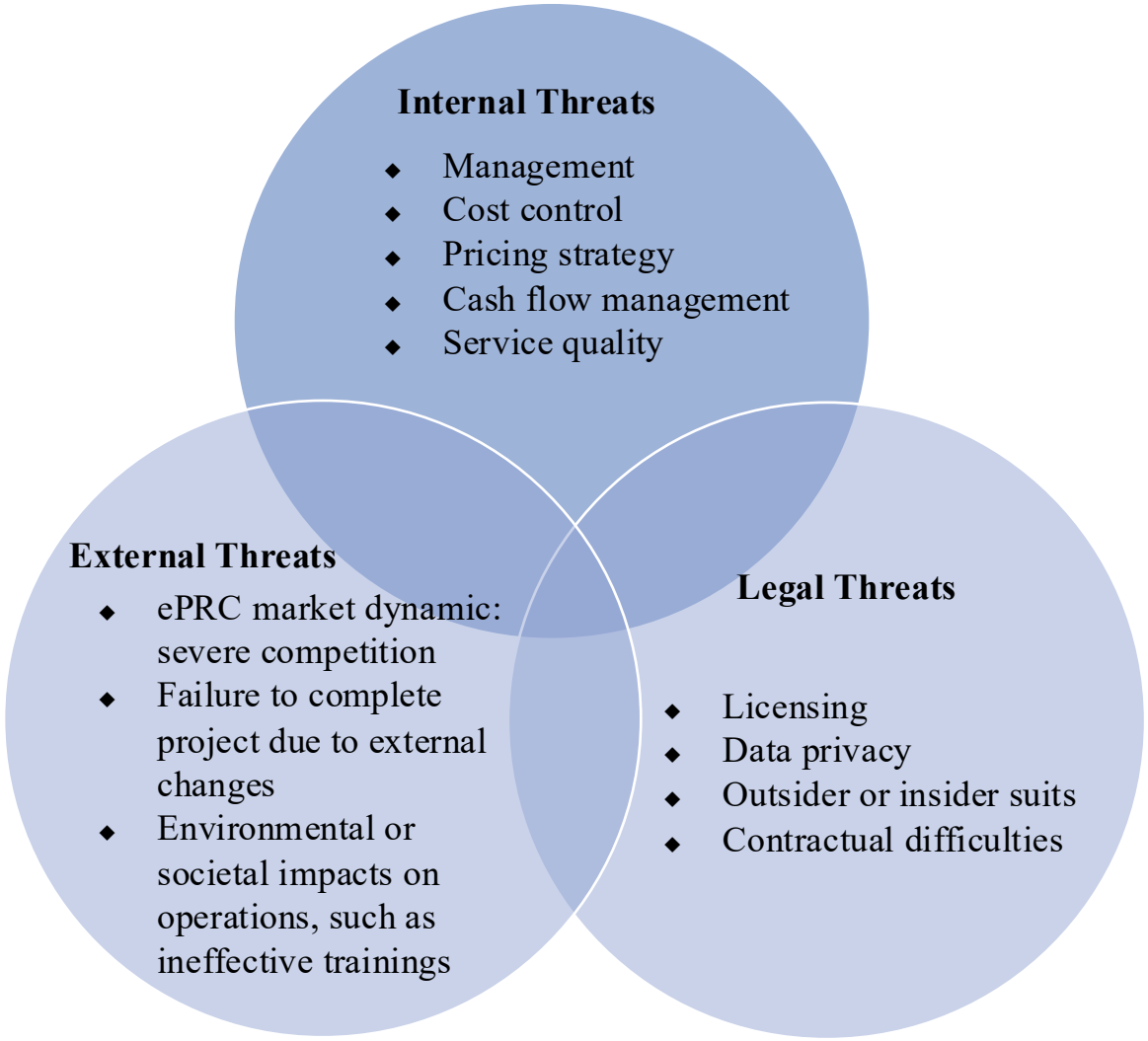
- **Size > 6**: 8.97% of the agencies, capturing 36.08% of the market
- **High efficiency** to break into the market

Geographic Type: Urban

- **Urban:** **44% of EMS agencies** in the US, meaning less implementation and communication costs

Grow Organically

We estimated the risks Company B is potentially facing from the perception of internal, external and legal threats, and recognized it's a good time for Company B to enter into the market



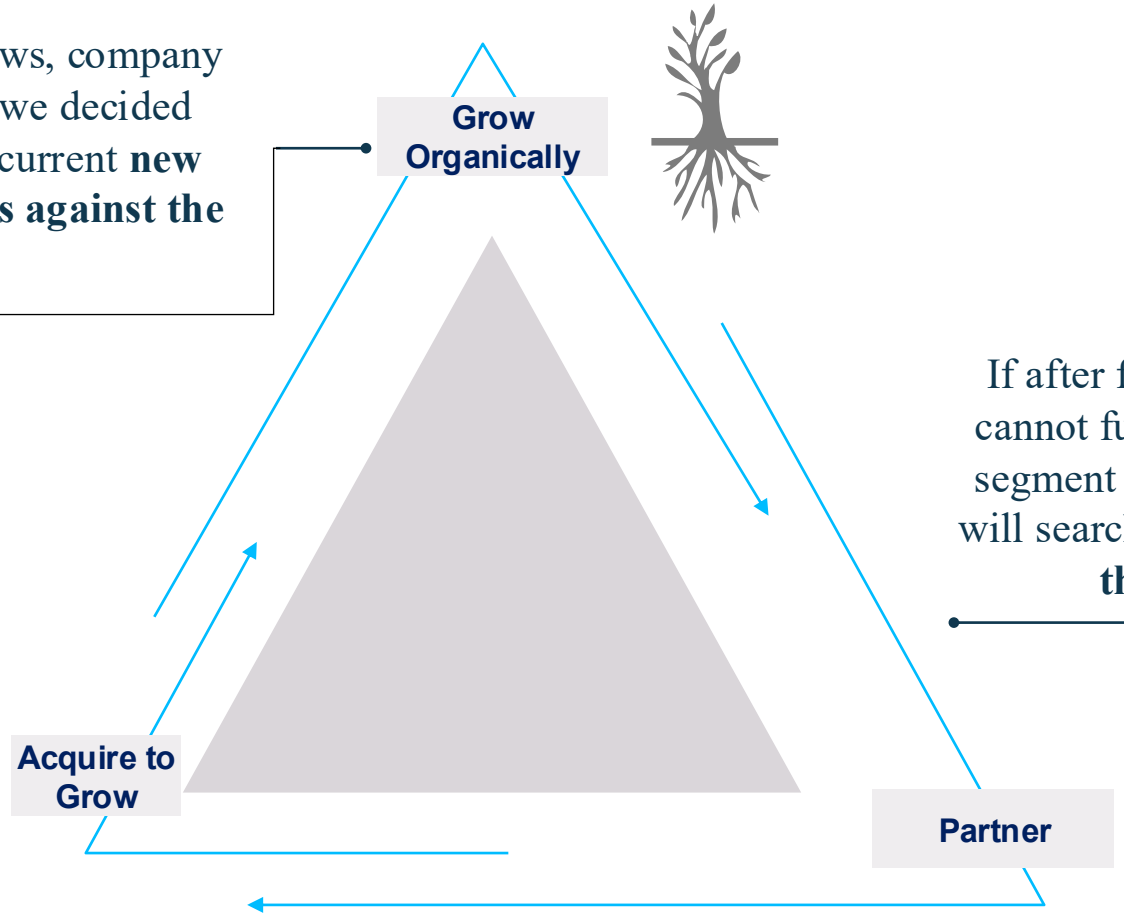
Conclusions & Impact on Company B

- Recognizing the potential risks, Company B should **seize the current market trend and be a fast mover**
- Both the **management board and staff** on this project must be aligned to execute, and be aware of the risks and challenges

Feasible to grow organically, but open to acquisition or JV in the future to accelerate market expansion

Utilizing results through interviews, company information and customer needs we decided that company B can utilize their current **new products to leverage themselves against the competition**.

Through research we may find value-add **acquisitions** that could help company B grow and **ultimately acquire additional market share or future revenues**



If after full analysis we feel that Company B cannot fully own market share in the targeted segment and continue to grow as a leader, we will search out and find **possible partners for them to utilize full market potential**

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Appendix

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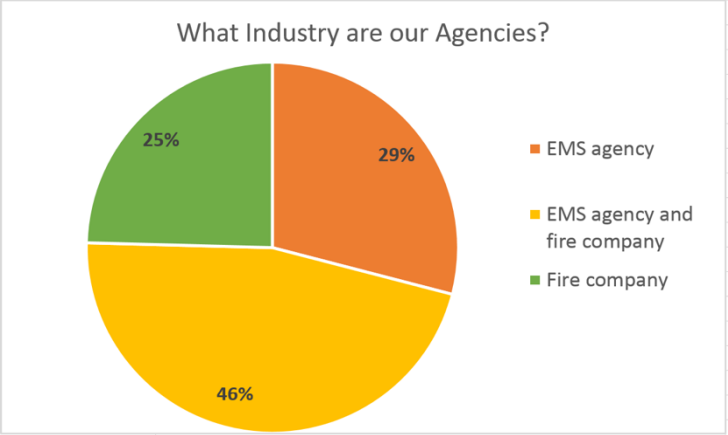
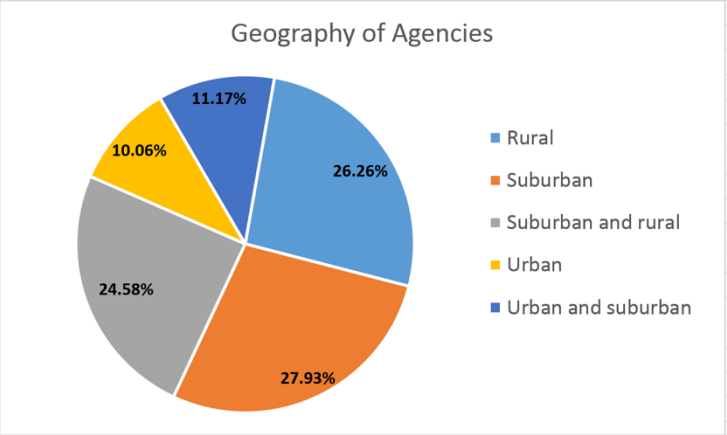
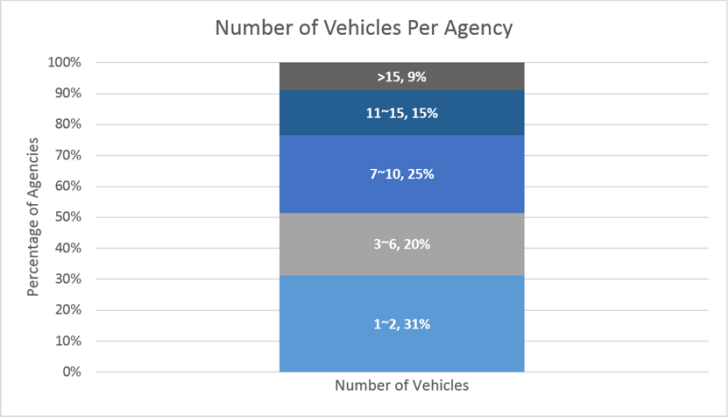
Nearly 80% of the agencies are located in rural or suburban area

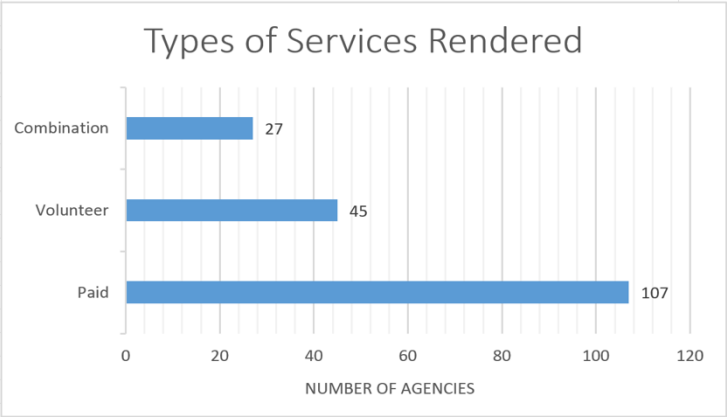
Agencies Attributes, Geography, Capacity and Services Rendered

From: Survey Data







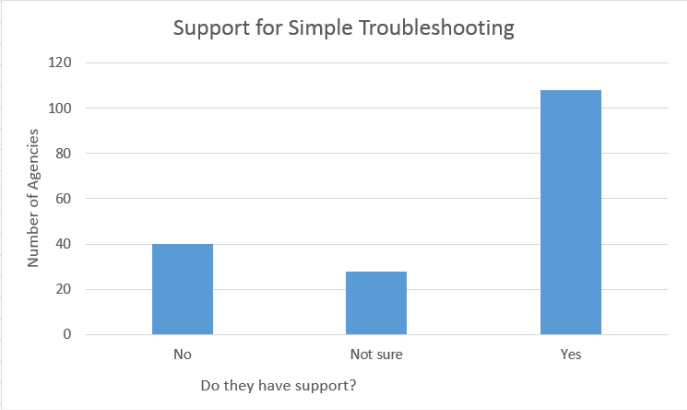


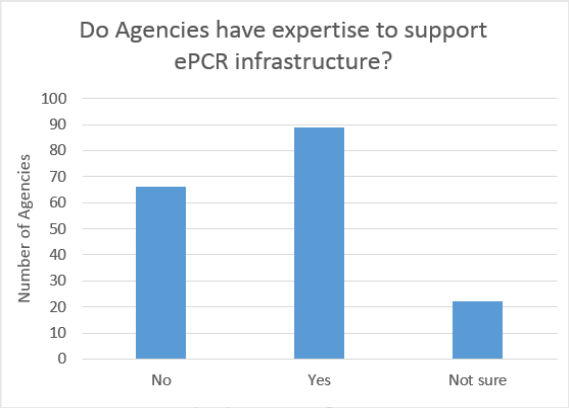
Observations

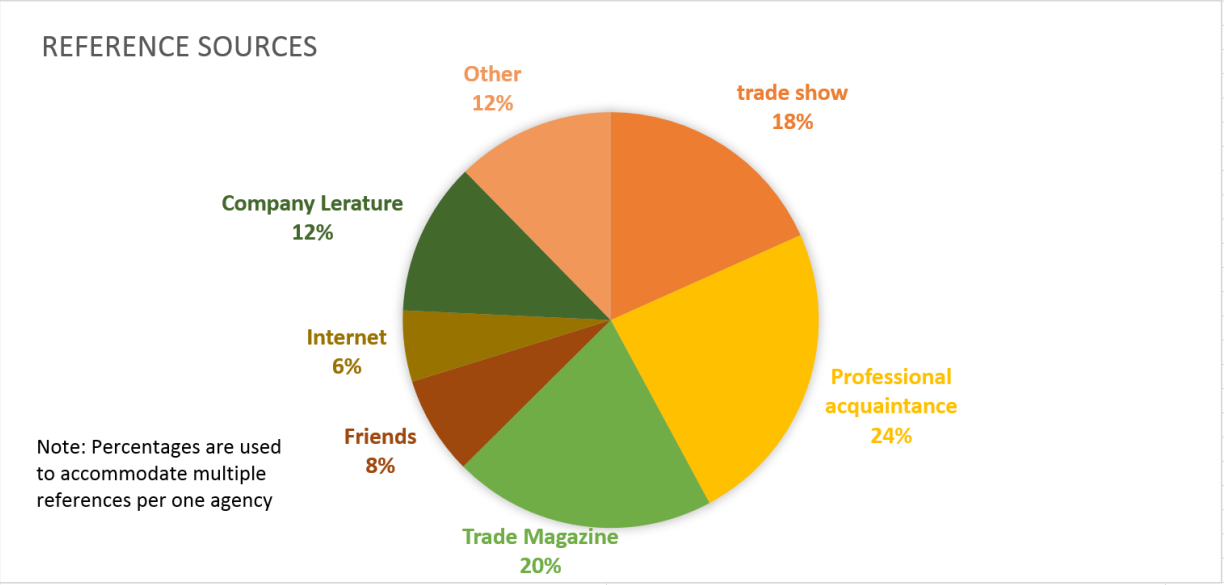
- Nearly half of the agencies are **EMS agencies** as well as **fire companies**.
- **Nearly 80%** of the agencies are located in rural or suburban area.
- **Only 24%** of the agencies have more than 10 ambulance vehicles.
- Agencies that provide paid services are **nearly 3 times** more than the agencies that offer free services

1. Customer Analysis – Sales Potential

A Substantial Amount of Agencies Have the Potential to Implement the System





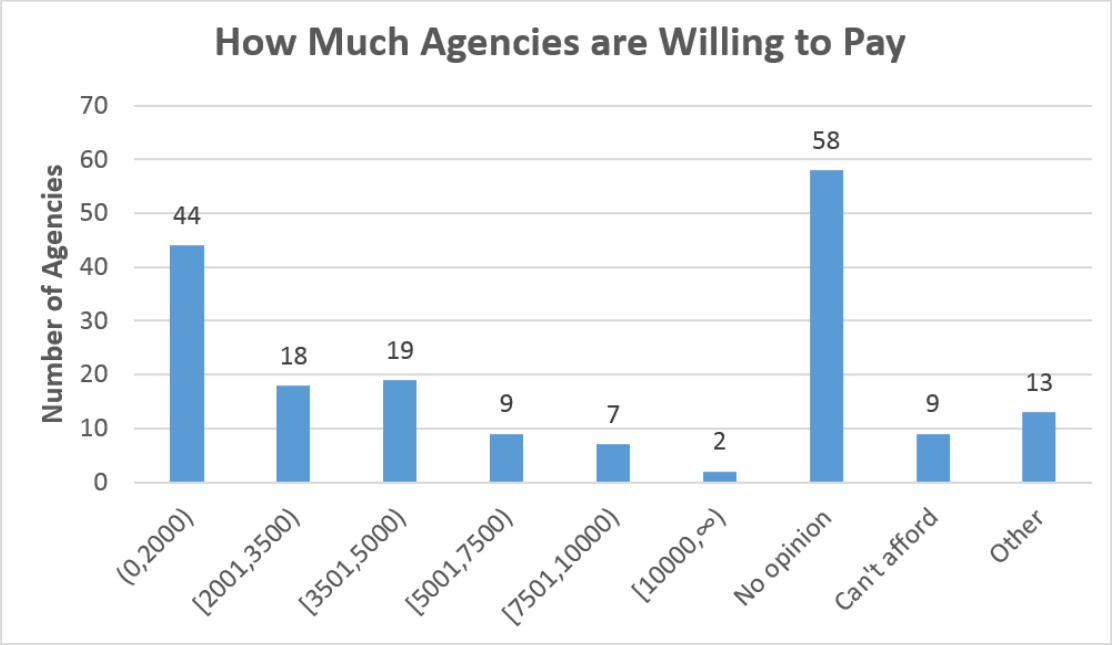


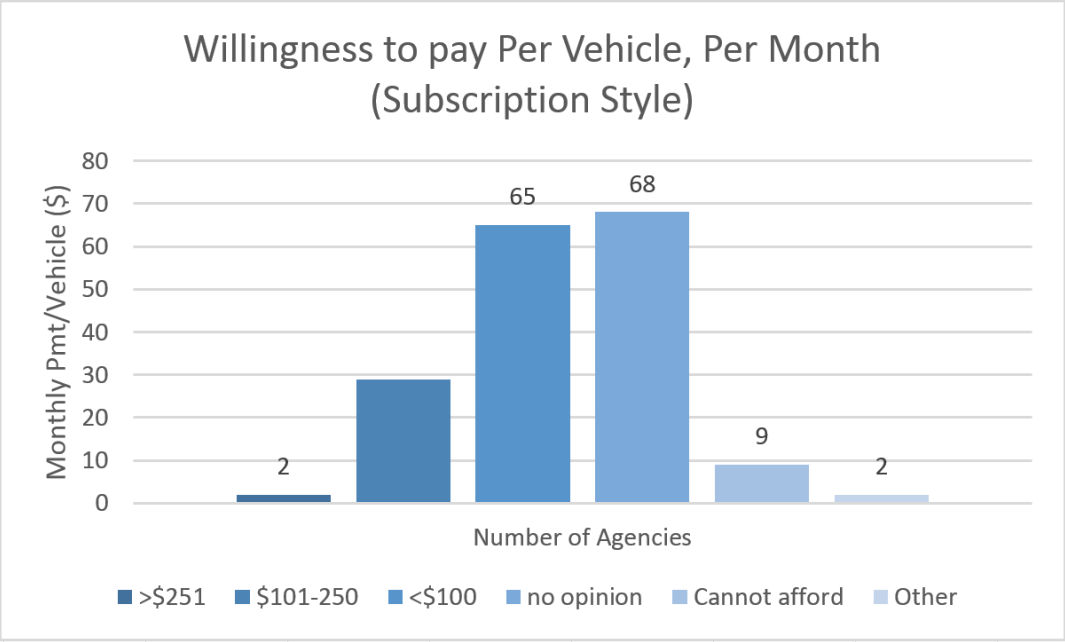
Conclusion

- Most agencies have on-site support for simple troubleshooting which offers more space for **system implementation**
- Half of the agencies **already have expertise** to support ePCR infrastructure
- Professional acquitance, trade magzine, and trade show are the **most common channels** for agencies to know about patient care record system

1. Customer Analysis – Willingness to Pay

Pricing Strategy Must be Adjusted to Meet Potential Customers' Expectation to Pay





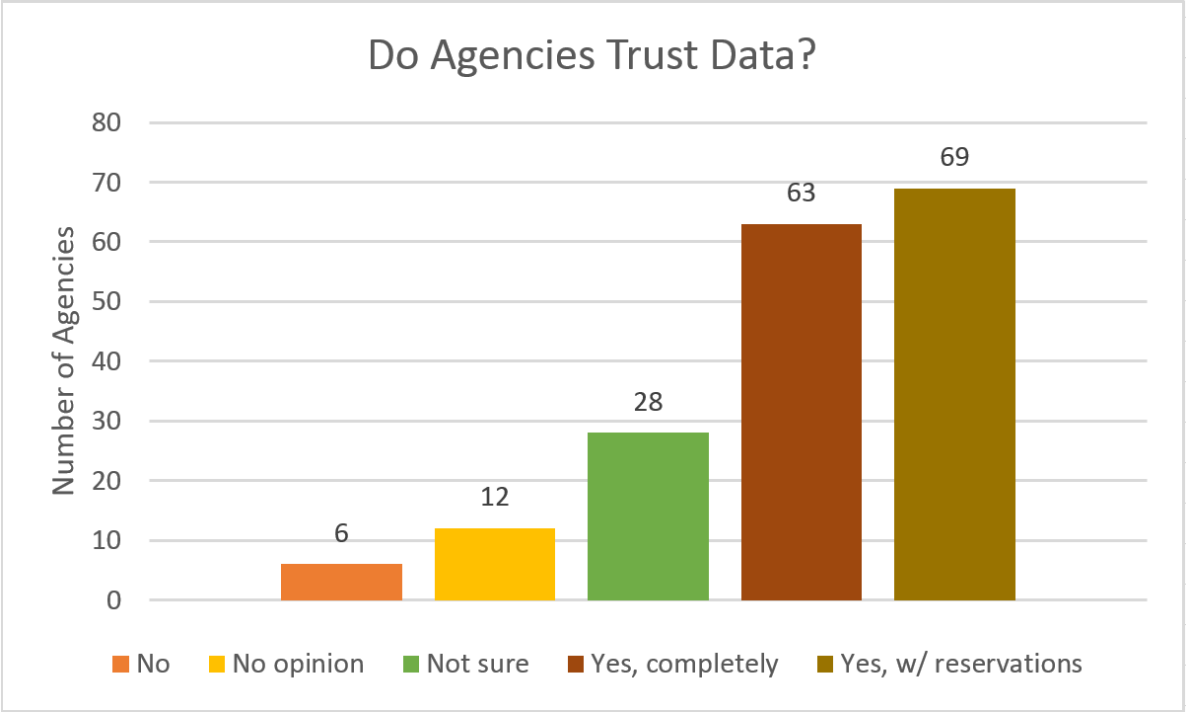
Survey Result of Agencies' Willingness to Pay for Patient Care Record System

- **Most agencies do not hold a clear opinion about paying for the system, or they are only willing to pay a relatively small amount**
- **If we choose a subscription style of payment, nearly half of the agencies would only expect to pay a small amount. Another case would be it takes effort to educate the market.**

1. Customer Analysis – Customer Relationship

Company B Will Face Challenges in Terms of Building up Customer Relationship

Survey Result of If Agencies Are Trusting Data

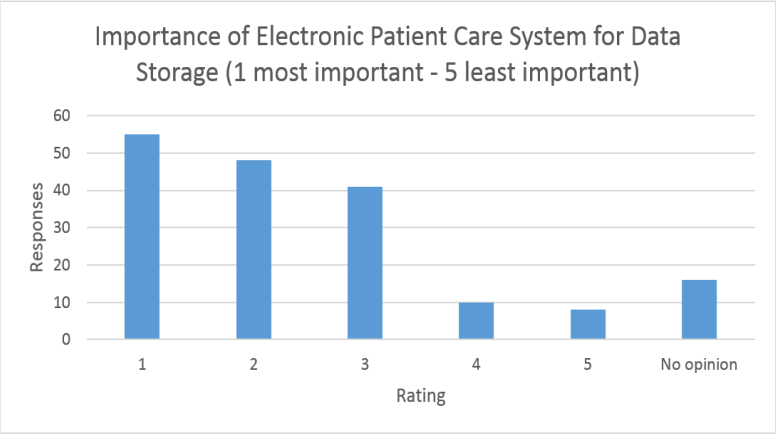


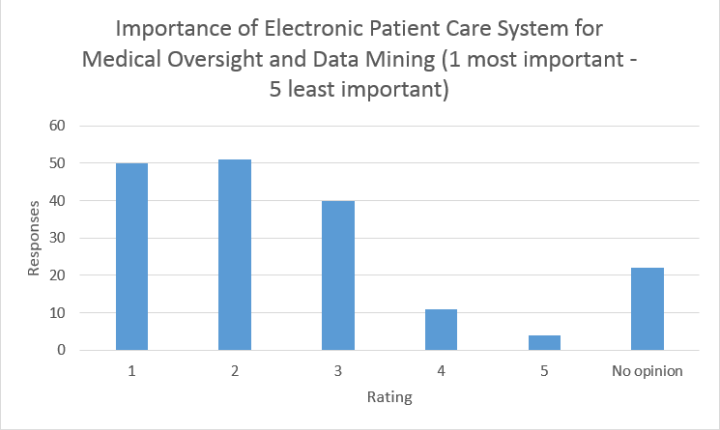
Conclusion

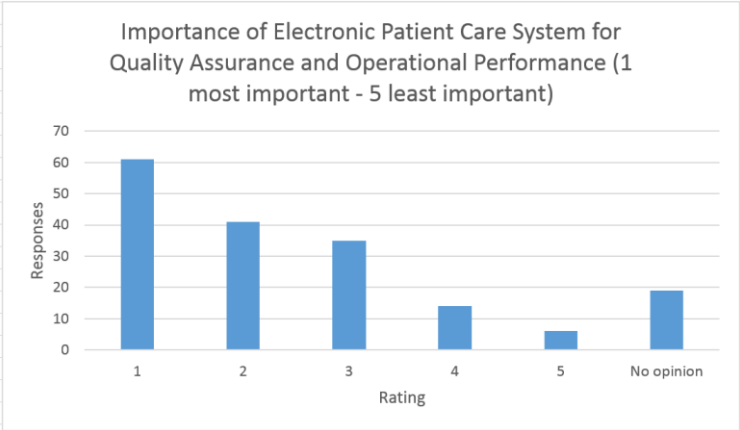
- A positive percentage of agencies are **completely supporting** the use of data and fully trust it.
- Meanwhile, slightly more amount of agencies are **holding doubts** on the authenticity of data.
- We need to be aware of the connection of their trust on data and **future relationship** with them, and it will effect their **willingness to pay** for the system.

2. Value Proposition – Technology & Quality

Agencies Value Functionalities such as Data Management, Medical Oversight and Data Mining





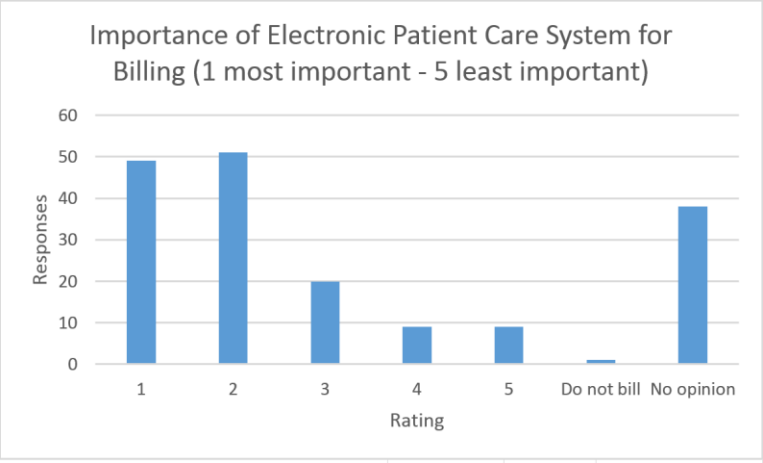


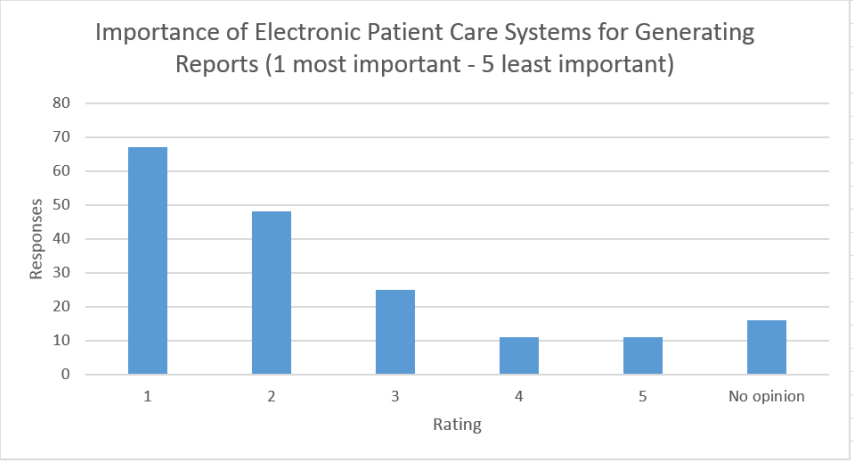
Survery Result of What Functionality Agencies Currently Value

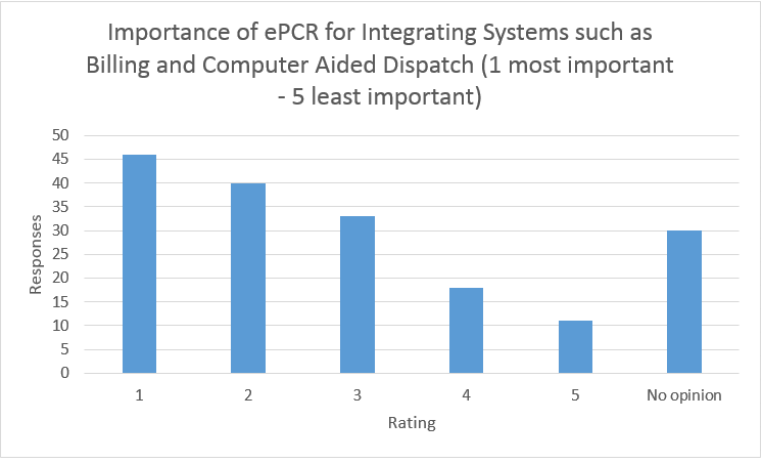
- **Most agencies rated the importance of data storage function of 1 to 3 out of 5**
- **Compared to data storage, more agencies do not have clear clue of medical oversight and data mining**
- **Quality assurance and operational performance should not be a priority for company B to develop**

2. Value Proposition – Technology & Quality (Cont.)

Company B Should Apply the System as an Integrating System with Diverse Functionalities







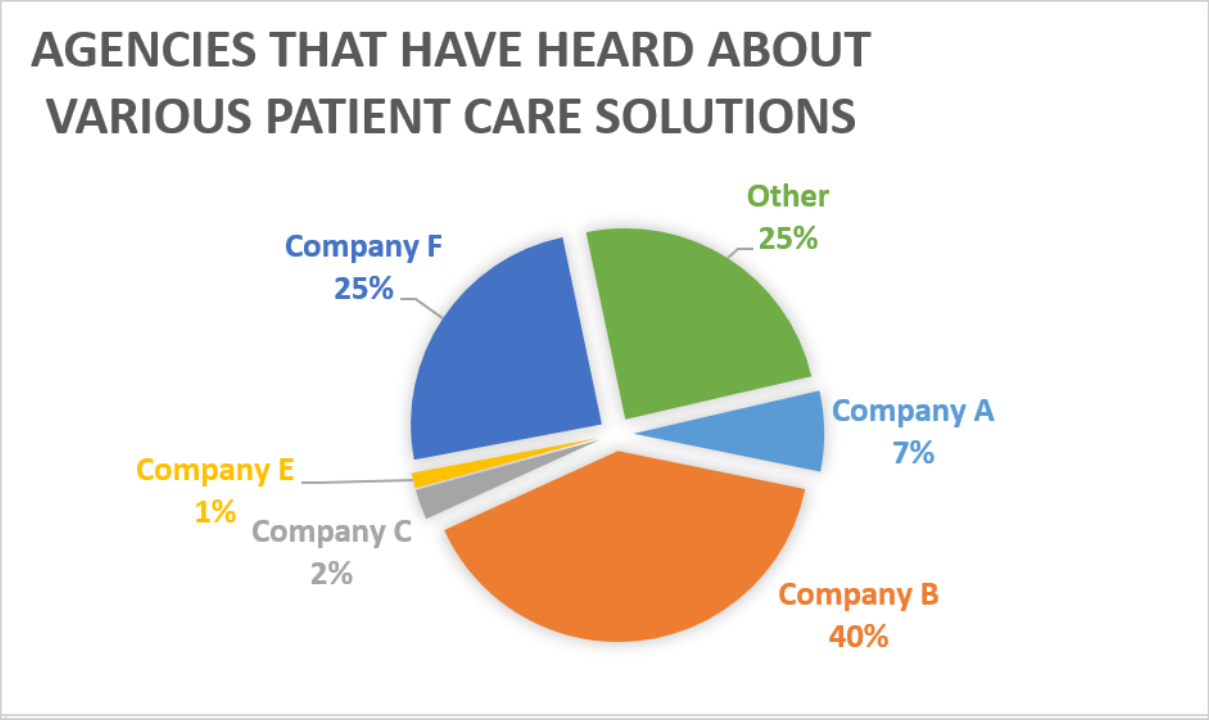
Survery Result of What Functionality Agencies Currently Value

- **Agencies are expecting to use electronic care system to collect payment from patients, while quite some of them are still observing**
- **Generating reports should not be a priority for Company B to develop on the early stage**
- **Many agencies expect ePCR system to have various functionalities to improve their operation efficiency**

3. Competitor Analysis

Company B Currently has a Dominant Position in the Industry Among Competitors

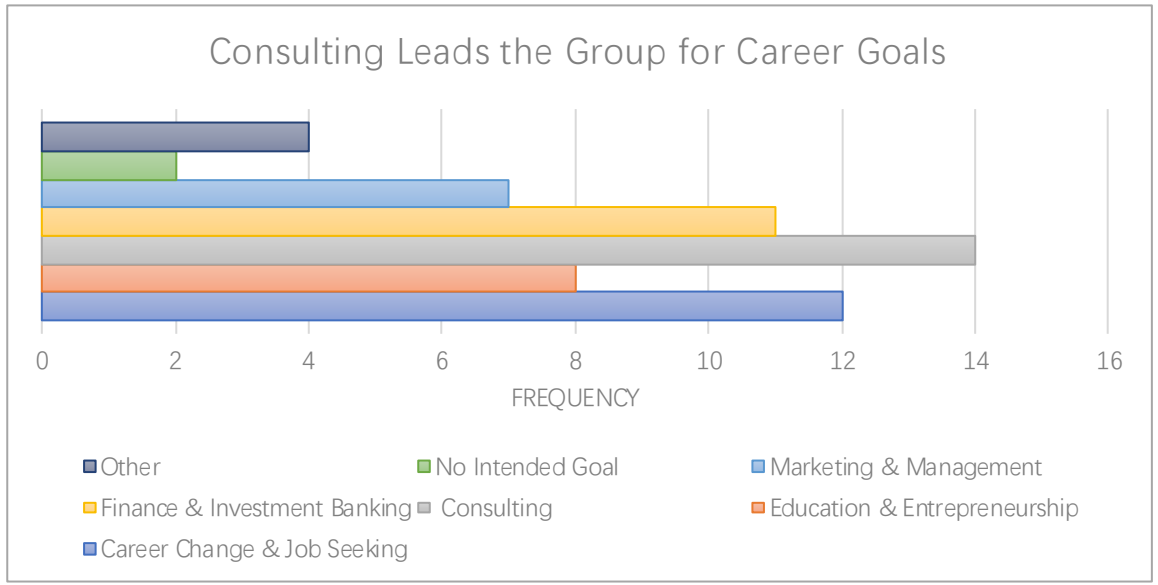
Survey Result of the Popularity of Patient Care Solution Providers

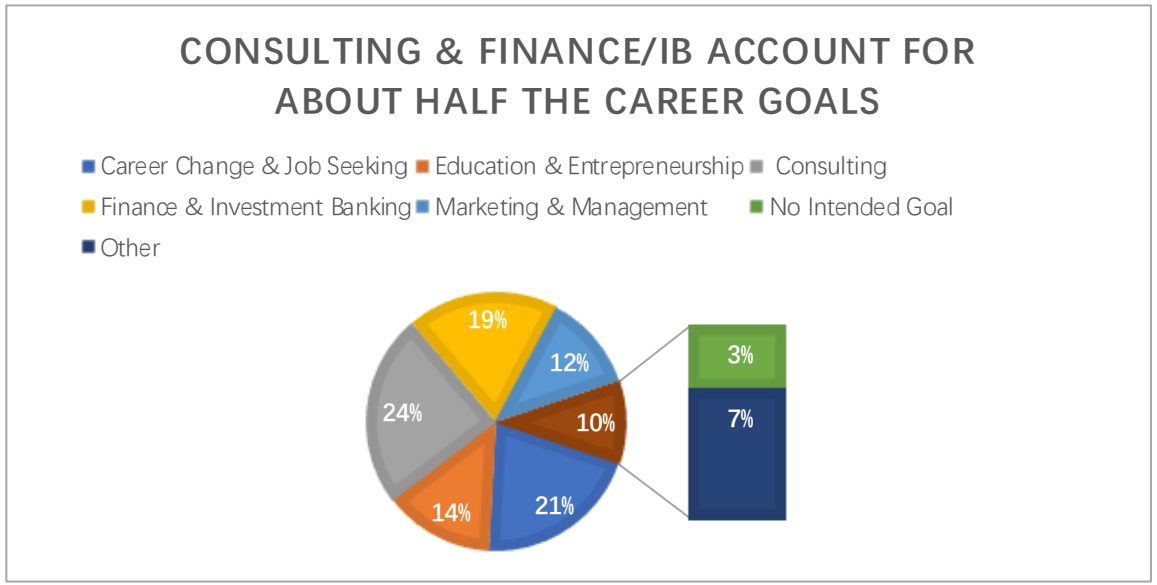


Conclusion

- Company B and company F are **the major players** to provide electronic patient care system
- Although company B has established a leading position in the industry, we still need to be aware of **company F's market share and future growth**

The Majority of Students prefer to go into Consulting or IB/Finance Related Fields





Conclusions & Clarifications

- **Consulting, followed by business account for 45% of students desired fields**
- **Marketing & Management and Education & Entrepreneurship remain on the same comparative level**
- **The other category consists of a combination of hospitality, location, sustainability, reinsurance**

EMS Agencies in the US Broken out by Geographic Type and Number of Vehicles

Geographic Type	Number of Agencies
Urban	10819
Suburban	2391
Rural	4950
Wilderness	2898
No Geo-Location Given	225

Number of Vehicles	Number of Agencies
1--2	6531
3--6	4199
7--10	5015
10--15	3266
>15	1866

Data Resource

- **Data was taken from the 2011 National EMS Assessment. Overall geographic characteristics were assumed to be similar to that of the inter-facility transfer agreement survey on page 517. Vehicle data was pulled from the question 2 survey data. From there, the percentages were multiplied by an extrapolated total of 21,283 EMS Agencies based on the survey results on page 21.**
- **Geographic Type: eg. Urban 1344/(1344+297+615+360+28)*20877 = 10819**
- **Number of Vehicles: eg. >15 Vehicles 16/(16+56+28+36+43)*20877 = 1866**
- **Resource: https://www.ems.gov/pdf/2011/National_EMS_Assessment_Final_Draft_12202011.pdf**

1--2	76.93% *19437	=14952
3--6	14.10% *19437	=2741
7--10	4.49% *19437	=872
11--15	2.56% *19437	=498
>15	1.92% *19437	=374

Rural and Wilderness	43.37% *19437	=8430
Suburban	12.25% *19437	=2381
Urban	44.37% *19437	=8625

Percentages based on NYS data by county. "Listing of Ambulance and Advanced Life Support First Response Services in New York " 2015

Total number of agencies comes from the National Association of State EMS Officials, "National EMS assessment 2011"

Percentages are based off of geographical breakdown survey for 800 EMS centers. "Characterizing Local EMS Systems", U.S. department of transportation, National Highway Traffic safety Administration, 2013.

Total number of agencies comes from the National Association of State EMS Officials, "National EMS assessment 2011"

Subscription Style

Avg(1-2)	1.5 *14952	=22428	
Avg(3-6)	4.5 *2741	=12334.5	
Avg(7-10)	8.5 *872	=7412	
Avg(11-15)	12.5 *498	=6225	
>15	16 *374	=5984	
Total Vehicle Assumption			54383.5
Fee Assumption (based on survey)		$	100.00
Months in a year			12
Annual Revenue if every vehicle subscribed to service		**$**	**65,260,200**

One time Fee

Avg(1-2)	1.5 *14952	=22428	
Avg(3-6)	4.5 *2741	=12334.5	
Avg(7-10)	8.5 *872	=7412	
Avg(11-15)	12.5 *498	=6225	
>15	16 *374	=5984	
Total Vehicle Assumption			54383.5
Fee Assumption (based on survey)		$	2,000.00
Revenue if every vehicle bought service		**$**	**108,767,000**

Row Labels	Count of How much would your agency be willing to pay per vehicle for such a system?
Response	**1**
Response	1
Rural	**47**
$2001 ~ $3500	1
$3,501-$5,000	1
$3501 ~ $5000	4
$5001 ~ $7500	1
<$2000	12
No opinion	20
None- can't afford	1
Other (please specify)	7
Suburban	**49**
$2001 ~ $3500	3
$3501 ~ $5000	6
$5001 ~ $7500	3
$7501 ~ $10000	2
<$2,000	1
<$2000	9
2001 ~ 2500	1
No opinion	19
Other (please specify)	5
(blank)	
Suburban and rural	**44**
$2001 ~ $3500	8
$3501 ~ $5000	6
$5001 ~ $7500	3
$7501 ~ $10000	2
<$2000	11
>$10000	1
No opinion	9
other	1
Other (please specify)	3
Urban	**18**
$2001 ~ $3500	1
$3501 ~ $5000	1
$5,001-7,500	1
$7501 ~ $10,000	1
$7501 ~ $10000	1
<$2000	7
>$10000	1
No opinion	4
Other (please specify)	1
Urban and suburban	**19**
$2001 ~ $3500	4
$3501 ~ $5000	1
$5001 ~ $7500	1
$7501 ~ $10000	1
<$2000	4
No opinion	6
Other (please specify)	2
(blank)	
Grand Total	**178**

Avg(1-2)	1.5	*14952	=22428	/54383.5	41.24%
Avg(3-6)	4.5	*2741	=12334.5	/54383.5	22.68%
Avg(7-10)	8.5	*872	=7412	/54383.5	13.63%
Avg(11-15)	12.5	*498	=6225	/54383.5	11.45%
>15	16	*374	=5984	/54383.5	11.00%

Total Vehicle Assumption 54383.5

Vehicles/Agency	Agencies	Vehicles	Ratio
1--2	76.93%	41.24%	0.54
3--6	14.10%	22.68%	1.61
7--10	4.49%	13.63%	3.04
11--15	2.56%	11.45%	4.47
>15	1.92%	11.00%	5.72